VPA IB DP CS — Topic 3: Networks (Weeks 1–9)

Topic 3: Networks 网络 Wǎngluò

IB Syllabus: Topic 3 (3.1–3.3) | Level: SL + HL | Duration: 9 weeks (30 hours)

Topic 3.1: Network Fundamentals 网络基础 Wǎngluò Jīchǔ

Week 1 — Introduction to Networks 网络简介 Wǎngluò Jiǎnjiè

Lesson	Content	Activities	Differentiation
1	Network types: LAN, WAN, PAN, WLAN, SAN, VPN (局域网, 广域网, 个人网)	Map VPA's network topology; identify network types	Bronze: Label network types; Silver: Explain differences; Gold: Analyse VPA network design decisions
2	Network topologies: bus, star, mesh, ring, tree, hybrid	Build physical models with string/cards; compare reliability	All: Build models; Gold: Calculate redundancy in mesh
3	Network hardware: NIC, switch, router, hub, bridge, gateway, access point (网卡, 交换机, 路由器)	Label physical hardware; trace packet path	Bronze: Identify devices; Silver: Explain function; Gold: Compare layer 2 vs layer 3 switches
4	Transmission media: twisted pair, coaxial, fibre optic, wireless; bandwidth and attenuation	Calculate file transfer times across different media	Bronze: Match media to use; Silver: Calculate; Gold: Discuss attenuation vs interference
5	TOK: Is the internet a single "thing," or is it better understood as an emergent property of countless interactions?	Socratic discussion; written reflection	All: Contribute; Gold: Reference complexity theory

TOK Knowledge Question: We speak of "the internet" as if it were a unified entity, yet it is a decentralised collection of autonomous networks. When we claim to "know" something we read online, how does the distributed, ungoverned nature of the network affect the reliability of that knowledge?

Week 2 — Protocols and Packet Switching 协议与分组交换 Xiéyì yǔ Fēnzǔ Jiāohuàn

Lesson	Content	Activities
1	Protocols: TCP, UDP, IP, HTTP, HTTPS, FTP, SMTP, POP3, IMAP (传输控制协议, 用户数据报协议, 网际协议)	Protocol matching game; trace an email's journey
2	The TCP/IP model: application, transport, internet, network access layers	Map OSI to TCP/IP; place protocols at correct layers
3	IP addressing: IPv4 structure, subnet mask, private vs public addresses, IPv6 necessity	Calculate network ranges; debate IPv6 adoption

Lesson	Content	Activities
4	Packet switching: packets, headers, routing, reassembly; circuit switching comparison	Simulation: students act as packets navigating a network
5	Formative: Protocols and packet switching quiz (25 marks)	Timed quiz; peer review

Week 3 — Network Utilities and Troubleshooting

Lesson	Content	Activities
1	Common network utilities: ping, traceroute, ipconfig/ifconfig, nslookup	Lab: use utilities to diagnose network issues
2	DNS: domain name system, resolution process, caching, root servers	Trace DNS lookup; discuss DNS security
3	DHCP: dynamic host configuration, lease time, IP allocation	Diagram DHCP handshake; configure a simple DHCP scope
4	MAC addressing vs IP addressing; ARP protocol	Packet capture observation (if permitted); ARP table inspection
5	Consolidation: Network troubleshooting case study	Groups diagnose and resolve a network fault scenario

Topic 3.2: Network Structures 网络结构 Wǎngluò Jiégòu

Week 4 — Client-Server and Peer-to-Peer 客户机-服务器与对等网 Kèhùjī-Fúwùqì yǔ Duìděng Wǎng

Lesson	Content	Activities
1	Client-server model: thin client, thick client, server roles (file, web, database, mail)	Diagram VPA's client-server infrastructure
2	Advantages and disadvantages of client-server	Debate: client-server vs alternative models
3	Peer-to-peer (P2P) networks: structure, use cases, BitTorrent, blockchain introduction	Research: legitimate and illegitimate P2P uses
4	Comparing C-S and P2P: cost, security, scalability, administration	Comparison table with justified conclusions
5	Formative: 8-mark "Compare" question on C-S vs P2P	Timed writing; peer mark with markscheme

Week 5 — Cloud and Virtual Networks 云与虚拟网络 Yún yǔ Xūnǐ Wǎngluò

Lesson	Content	Activities
1	Cloud computing models: IaaS, PaaS, SaaS (基础设施即服务, 平台即服务, 软件即服务)	Classify VPA's cloud services; calculate cost comparisons

Lesson	Content	Activities
2	Public, private, hybrid cloud; advantages and risks	Evaluate cloud migration for a given organisation
3	Virtual private networks (VPN): tunneling, encryption, use cases	Diagram VPN operation; discuss censorship circumvention ethics
4	Virtual LANs (VLAN), software-defined networking (SDN) — HL extension	Design VLANs for a multi-department school
5	TOK: When your data is "in the cloud," where is it really? Who owns it? Who can access it?	Ethical discussion; connection to knowledge claims

TOK Knowledge Question: Cloud storage blurs the line between possession and access. If you store your photos "in the cloud," do you know where they are? Does physical location matter for ownership, privacy, and legal jurisdiction? How does trust replace direct knowledge in networked systems?

Topic 3.3: Network Security 网络安全 Wǎngluò □nquán

Week 6 — Threats and Vulnerabilities 威胁与漏洞 Wēixié yǔ Lòudòng

Lesson	Content	Activities
1	Security threats: malware (virus, worm, trojan, ransomware, spyware), phishing, DDoS, MITM (恶意软件, 钓鱼, 拒绝服务攻击)	Research recent major breaches; create threat awareness poster
2	Vulnerabilities: unpatched software, weak passwords, social engineering, insider threats	Password strength analysis; social engineering scenario
3	Authentication: something you know, have, are; MFA, biometrics	Evaluate authentication for a banking app
4	Access control: MAC, DAC, RBAC (强制/自主/基于角色的访问控制)	Design RBAC for a hospital system
5	Formative: Security scenario analysis (identify threats, recommend controls)	Case study; structured response

Week 7 — Encryption and Protocols 加密与协议 Jiāmì yǔ Xiéyì

Lesson	Content	Activities
1	Encryption: symmetric (AES), asymmetric (RSA), hashing (SHA) (对称加密, 非对称加密, 哈希)	Caesar cipher → modern encryption progression
2	Public key infrastructure (PKI): certificates, CAs, trust chains	Examine a website's certificate; verify chain
3	SSL/TLS: handshake process, HTTPS, padlock meaning	Diagram SSL handshake; explain why it matters
4	Firewalls: packet filtering, stateful inspection, application layer, proxy	Design firewall rules for a school network

Lesson	Content	Activities
5	TOK: Does encryption protect knowledge, or does it hide it? When should governments have the right to decrypt?	Structured debate; preparation for Paper 1 essays

TOK Knowledge Question: Governments argue that "backdoors" in encryption are necessary for security; cryptographers argue they destroy security. Is this a dispute about facts, or about values? Can we know whether a backdoor will be exploited, or is this a matter of probabilistic judgement?

Week 8 — Topic 3 Consolidation

Lesson	Content	Activities
1	Revision: Network architecture and protocols mind map	Collaborative digital mind map
2	Revision: Security mechanisms and encryption review	Flashcard competition
3	Exam technique: network diagram questions (6 marks)	Guided practice with markscheme
4	Exam technique: 10-mark "Evaluate" questions on network security	Essay planning; peer review of plans
5	Formative: Topic 3 progress test (40 marks, Paper 1 style)	Timed; self-assessment

Week 9 — Topic 3 End-of-Topic Assessment

Lesson	Content	Activities
1	Revision clinic: student-requested topics	Student-led Q&A
2	Assessment: Topic 3 End-of-Topic Test (1 hour, 50 marks)	Formal conditions
3	Test feedback and DIRT	Markscheme analysis
4	Introduction to Topic 4: What is computational thinking?	Unplugged activity: sorting algorithms with students
5	Topic 4 preview: programming environments and pseudocode conventions	Set up IDE; write first pseudocode

Key Vocabulary (Topic 3)

English	中文	Pinyin	Definition
LAN	局域网	júyù wǎng	Local Area Network: covers a small geographical area
WAN	广域网	guǎngyù wǎng	Wide Area Network: spans large geographical areas
Protocol	协议	xiéyì	A set of rules governing data communication

English	中文	Pinyin	Definition
Packet switching	分组交换	fēnzǔ jiāohuàn	Breaking data into packets sent independently across a network
Router	路由器	lùyóu qì	A device that forwards packets between networks
IP address	IP地址	IP dìzhǐ	A unique numerical identifier for a device on a network
DNS	域名系统	yùmíng xìtǒng	Translates domain names to IP addresses
Encryption	加密	jiāmì	Converting data into a coded form to prevent unauthorised access
Firewall	防火墙	fánghuǒqiáng	A security system monitoring and controlling network traffic
VPN	虚拟专用网络	xūnǐ zhuānyòng wǎngluò	Extends a private network across a public network securely

Teacher Checklist — End of Topic 3:

- ☐ All IB 3.1–3.3 syllabus points covered
- ☐ Network simulation or practical activities completed
- ☐ TOK discussions documented
- ☐ EAL vocabulary tracked
- ☐ IA client meetings ongoing
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Topic 3 Scheme of Work